Digital Image Processing Lab Assignment 1

Objective

- To understand how images are represented using arrays.
- To learn grayscale and RGB image construction.
- To analyze pixel intensity differences.

Task 1: Grayscale Image from Array

Instructions:

1. Create a 2D array (matrix) with values ranging from 0 to 255.
2. Convert this array into a grayscale image.
3. Display the image.
4. Observe how different values in the array affect the brightness (intensity) of the image.
5. Convert the generated grayscale image back into an array.

Task 2: RGB Image Construction (Half Green, Half Red)

Instructions:

1. Create a 3D array representing an RGB image.

   Shape: (height × width × 3)

2. Divide the image into two equal halves:

   Left half → Red color (R=255, G=0, B=0)

   Right half → Green color (R=0, G=255, B=0)

3. Convert the array into an image and display it.

Submission Requirements

- Python code for both tasks.
- Output images.

Tools/Libraries Suggested

- NumPy
- OpenCV or PILLOW